



HUNKAARTM 2026

TOPIC

→ Integration :-

अध्याय – 7 समाकलन



7.3 → Completed

परिचय

7.4 →

Q- $\int \operatorname{cosec} x \, dx$

(A) $\log |\sin x| + c$

(B) $\log |\operatorname{cosec} x - \cot x| + c$ ✓

(C) $\log |\sec x - \tan x| + c$

(D) $-\log |\cos x| + c$

Q- $\int \tan x \, dx = \log |\sec x| + c$ or $-\log |\cos x| + c$

Q- $\int \sec x \, dx = \log |\sec x + \tan x| + c$

परिचय

7.4 →

कुछ विशिष्ट प्रकार के फलनों का समाकलन
(Some particular type of function)

$$(i) \int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$$

$$(ii) \int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$(iii) \int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$$



Example ÷

$$Q - \int \frac{1}{x^2 - 16} dx$$

$$\Rightarrow \int \frac{1}{x^2 - 4^2} dx = \int \frac{1}{x^2 - a^2} dx$$

$a = 4$

$$= \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$$

$$\Rightarrow \boxed{\frac{1}{8} \log \left| \frac{x-4}{x+4} \right| + C} \text{ Ans}$$

$$Q - \int \frac{1}{x^2 + 16} dx \quad \boxed{a=4}$$

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$\boxed{\frac{1}{4} \tan^{-1} \frac{x}{4} + C} \text{ Ans}$$

$$Q - \int \frac{1}{16 - x^2} dx \quad a=4$$

$$\int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$$

$$\Rightarrow \frac{1}{2 \times 4} \log \left| \frac{4+x}{4-x} \right| + C$$

$$\Rightarrow \boxed{\frac{1}{8} \log \left| \frac{4+x}{4-x} \right| + C}$$

परिचय

$$(4) \int \frac{1}{\sqrt{x^2 - a^2}} dx = \log |x + \sqrt{x^2 - a^2}| + C$$

$$(5) \int \frac{1}{\sqrt{x^2 + a^2}} dx = \log |x + \sqrt{x^2 + a^2}| + C$$

$$(6) \int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + C$$

Q- $\int \frac{dx}{\sqrt{x^2-16}}$

$$\int \frac{1}{\sqrt{x^2-a^2}} = \log|x + \sqrt{x^2-a^2}| + C$$

$$\log|x + \sqrt{x^2-4^2}| + C$$

Q- $\int \frac{dx}{\sqrt{x^2+16}}$

$$\int \frac{1}{x^2+a^2} = \log|x + \sqrt{x^2+a^2}| + C$$

$$\log|x + \sqrt{x^2+4^2}| + C$$

Q- $\int \frac{dx}{\sqrt{16-x^2}}$

$$\int \frac{1}{a^2-x^2} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$\Rightarrow \sin^{-1}\left(\frac{x}{4}\right) + C$$

⑦ $\int \frac{\sqrt{x}}{ax^2+bx+c}$

Step-1 ÷ Check it - x^2 का गुणांक क्या 1 है। - (a)

Step-2 ÷ x के गुणांक की आशा करना तथा समीकरण बना करके जोड़ तथा घटा देना है।

Step-3 ÷ हल कर देना है।

⑧ $\int \frac{\sqrt{x}}{\sqrt{ax^2+bx+c}}$

→ Same

$$(g) \quad \int \frac{px+q}{ax^2+bx+c} dx = A \frac{d}{dx} (ax^2+bx+c) + B$$

Q-

$$\int \frac{\sqrt{x}}{x^2 - 6x + 13}$$

n का गुणांक = -6

$$\frac{-b}{2} \Rightarrow (-3)$$

$$\Rightarrow \int \frac{\sqrt{x}}{x^2 - 6x + 13 + (-3)^2 - (-3)^2}$$

$$\Rightarrow \int \frac{\sqrt{x}}{x^2 - 6x + (-3)^2 + 13 - 9}$$

$n^2 - 2ab + b^2$

$\Rightarrow$

$$\int \frac{\sqrt{u}}{(u-3)^2 + 4}$$

$\Rightarrow$

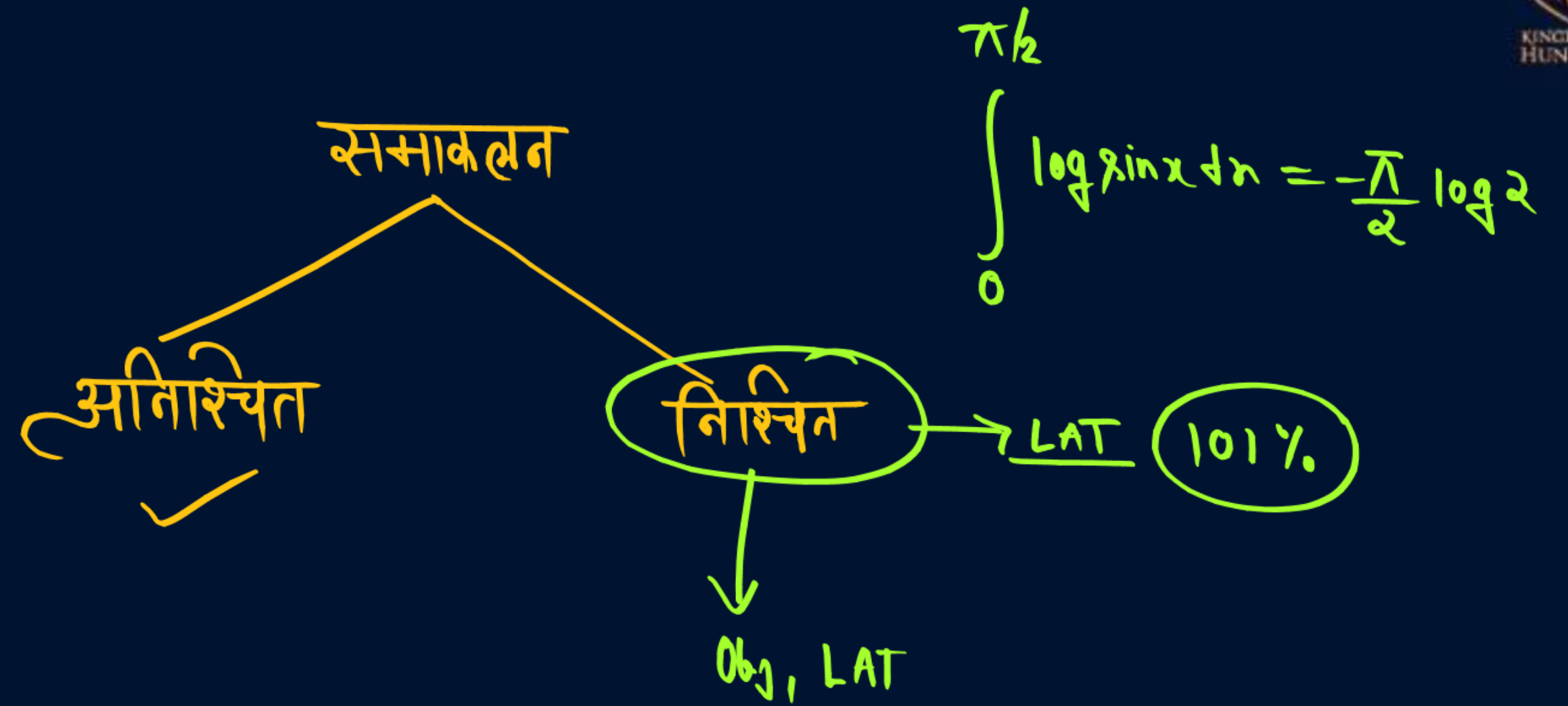
$$\int \frac{\sqrt{x}}{(x-3)^2 + 2^2} \rightarrow a=2$$

$x^2 + a^2$

$$\int \frac{1}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

$$\Rightarrow \boxed{\frac{1}{2} \tan^{-1} \left(\frac{x-3}{2} \right) + C}$$

Ans



7.4 → Introduction

- (1) There are 9 type of Question ÷
- (II) Formula ÷ 9

NEXT CLASS

7.4 → learn all 9 type of formula

one-shot → Definite

DAILY PRACTICE QUIZ